TADHG REID

MECHATRONIC ENGINEER

CONTACT

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WEBSITE

https://tadhg-reid.github.io

SKILLS

- First-principles calculations, concept and prototype development to ISO standards
- SolidWorks, PTC Creo
- BS-8888 GD&T drawings for traditional manufacturing
- ANSYS (stress, pressure, thermal, vibration)
- SLA, FDM, Thermoforming,
- Cleanroom and laboratory practice trained
- FRT surface metrology & CMM
- MATLAB, Scilab & Simulink
- C, C++, Java & Python (ML, image processing)

CERTIFICATIONS

- Full Irish Driving License (5 Yr)
- SolidWorks Design Associate (CSWA), Dassault Systèmes, 2022
- SOLAS Safe Pass
- Manual Handling Certificate

LANGUAGES

- English – Fluent
- Spanish – Intermediate

ADDITIONAL INFO

- Equipment Officer for DCU Engineering Society
- Enjoys repairing electronics and electric guitars
- Member of Engineers Ireland, aiming for Chartered Engineer

PROFESSIONAL SUMMARY

Mechatronic Engineering graduate with First Class Honours and hands-on experience in mechanical design, prototyping and validation across aerospace and medical-device sectors. Proficient in taking concepts from first-principles calculations through CAD, DFMEA, BS-8888 drawing packages, Ansys FEA and prototype testing. Strong understanding of DFM and engineering standards, with practical experience in cleanroom and laboratory environments. Aiming for a role in R&D, driven by a passion for continuous learning.

WORK EXPERIENCE

Research Assistant **05/2026 – Present**
RAPID, DCU Glasnevin

- Researched medical devices for applications including fluorescence microscopy, Lab on a Disc microfluidics and Lab on a Chip
- Designed and developed microfluidic discs, chips and other medical-device components in SolidWorks
- Rapid prototyping using stereolithography, thermoforming and FDM 3D printing for iterative design development
- Applied mechanical, electronic, software and chemical engineering to real-world medical-device projects

Aerospace Design Engineer Intern **03/2024 – 12/2024**
Nammo Ireland, Dublin

- Designed two concept valves up to preliminary prototypes for aerospace applications using mechanical design principles
- Developed designs from first principles in line with AS9100 and ISO standards, using Excel calculations to verify the design
- Drafted two complete BS-8888 GD&T drawing packages with DFM practices, in consultation with on-site machinists
- Validated designs through FEA for mechanical stress, pressure, temperature and vibration requirements
- Followed GMP and GDP to produce work instructions and DFMEAs
- Assembled and tested prototype valves in an ISO Class 9 cleanroom, gaining hands-on validation experience

EDUCATION

Masters, Mechatronic Engineering 05/2026
Dublin City University, Ireland

- First Class Honours – Rank One Student; *DCU Dean's Honours List Award 2026*

Bachelors, Mechatronic Engineering 05/2026
Dublin City University, Ireland

- First Class Honours – Rank One Student; *DCU Dean's Honours List Award 2026*